# Predictive Modeling Project

# Problem 1

You are hired by a company Gem Stones co ltd, which is a cubic zirconia manufacturer. You are provided with the dataset containing the prices and other attributes of almost 27,000 cubic zirconia (which is an inexpensive diamond alternative with many of the same qualities as a diamond). The company is earning different profits on different prize slots. You have to help the company in predicting the price for the stone on the basis of the details given in the dataset so it can distinguish between higher profitable stones and lower profitable stones so as to have better profit share. Also, provide them with the best 5 attributes that are most important.

1.1. Read the data and do exploratory data analysis. Describe the data briefly. (Check the null values, Data types, shape, EDA). Perform Univariate and Bivariate Analysis.

```
In [2]: data = pd.read_csv('cubic_zirconia1.csv')
```

```
In [3]: data.head()
Out[3]:
```

| | 0 | carat | cut | color | clarity | depth | table | x | y | z | price |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0.30 | Ideal | E | SI1 | 62.1 | 58.0 | 4.27 | 4.29 | 2.66 | 499 |
| **1** | 2 | 0.33 | Premium | G | IF | 60.8 | 58.0 | 4.42 | 4.46 | 2.70 | 984 |
| **2** | 3 | 0.90 | Very Good | E | VVS2 | 62.2 | 60.0 | 6.04 | 6.12 | 3.78 | 6289 |
| **3** | 4 | 0.42 | Ideal | F | VS1 | 61.6 | 56.0 | 4.82 | 4.80 | 2.96 | 1082 |
| **4** | 5 | 0.31 | Ideal | F | VVS1 | 60.4 | 59.0 | 4.35 | 4.43 | 2.65 | 779 |

- Data has been read and checked using Head() function

# Summary and shape of the dataset

```
In [7]: data.shape
Out[7]: (26967, 10)

In [8]: data.describe()
Out[8]:
```

| | carat | depth | table | x | y | z | price |
|---|---|---|---|---|---|---|---|
| **count** | 26967.000000 | 26270.000000 | 26967.000000 | 26967.000000 | 26967.000000 | 26967.000000 | 26967.000000 |
| **mean** | 0.798375 | 61.745147 | 57.456080 | 5.729854 | 5.733569 | 3.538057 | 3939.518115 |
| **std** | 0.477745 | 1.412860 | 2.232068 | 1.128516 | 1.166058 | 0.720624 | 4024.864666 |
| **min** | 0.200000 | 50.800000 | 49.000000 | 0.000000 | 0.000000 | 0.000000 | 326.000000 |
| **25%** | 0.400000 | 61.000000 | 56.000000 | 4.710000 | 4.710000 | 2.900000 | 945.000000 |
| **50%** | 0.700000 | 61.800000 | 57.000000 | 5.690000 | 5.710000 | 3.520000 | 2375.000000 |
| **75%** | 1.050000 | 62.500000 | 59.000000 | 6.550000 | 6.540000 | 4.040000 | 5360.000000 |
| **max** | 4.500000 | 73.600000 | 79.000000 | 10.230000 | 58.900000 | 31.800000 | 18818.000000 |

# Information regarding dataset

```
In [6]: data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26967 entries, 0 to 26966
Data columns (total 10 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   carat    26967 non-null  float64
 1   cut      26967 non-null  object
 2   color    26967 non-null  object
 3   clarity  26967 non-null  object
 4   depth    26270 non-null  float64
 5   table    26967 non-null  float64
 6   x        26967 non-null  float64
 7   y        26967 non-null  float64
 8   z        26967 non-null  float64
 9   price    26967 non-null  int64
dtypes: float64(6), int64(1), object(3)
memory usage: 2.1+ MB
```

## Data Types

```
In [9]: data.dtypes

Out[9]: carat       float64
        cut          object
        color        object
        clarity      object
        depth       float64
        table       float64
        x           float64
        y           float64
        z           float64
        price         int64
        dtype: object
```

## Null value check

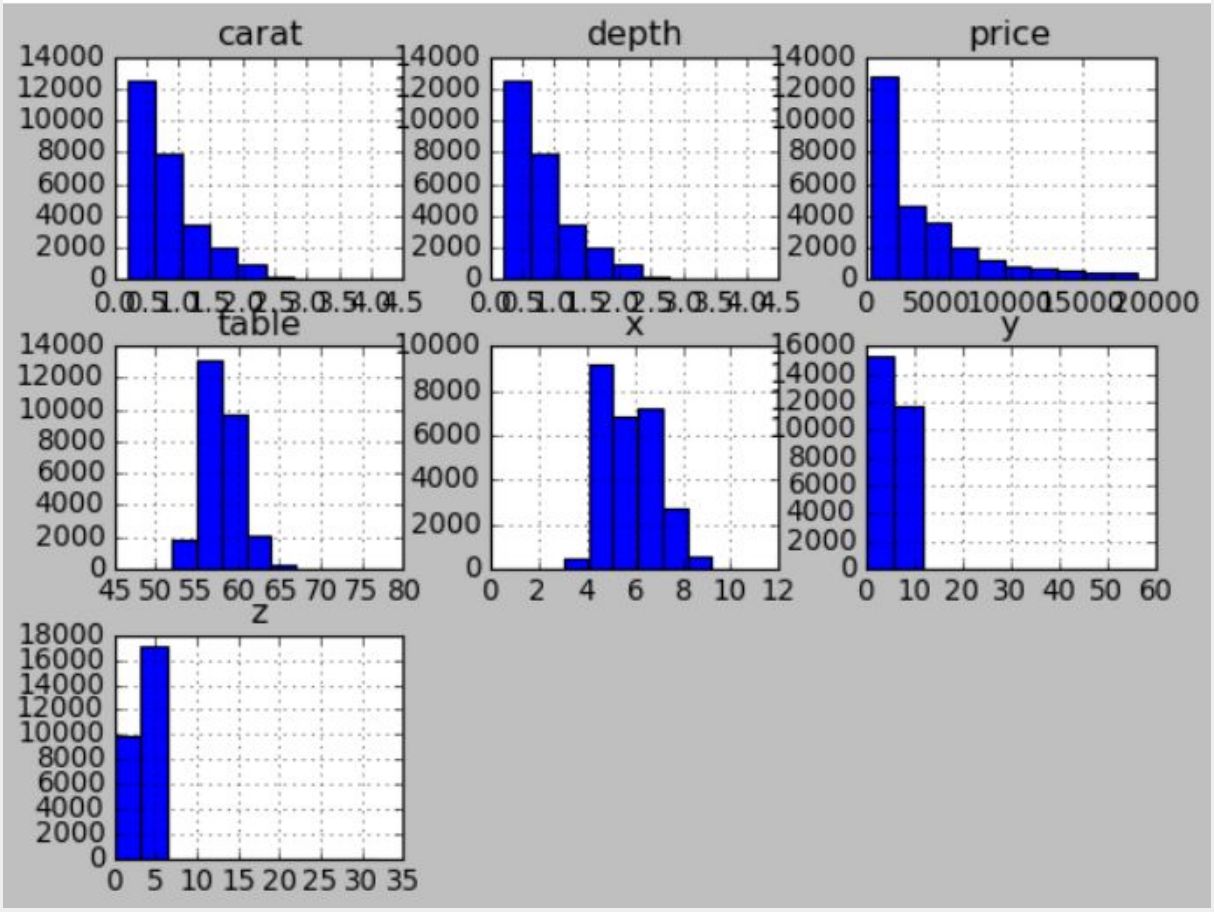
```
In [10]: data.isnull().sum()

Out[10]: carat        0
         cut          0
         color        0
         clarity      0
         depth      697
         table        0
         x            0
         y            0
         z            0
         price        0
         dtype: int64
```

- The dataset contains null values
- There are three object types, one integer type and six float type features in the dataset
- The dataset contains 26967 observations and 10 features
- Duplicates in the dataset have been dropped

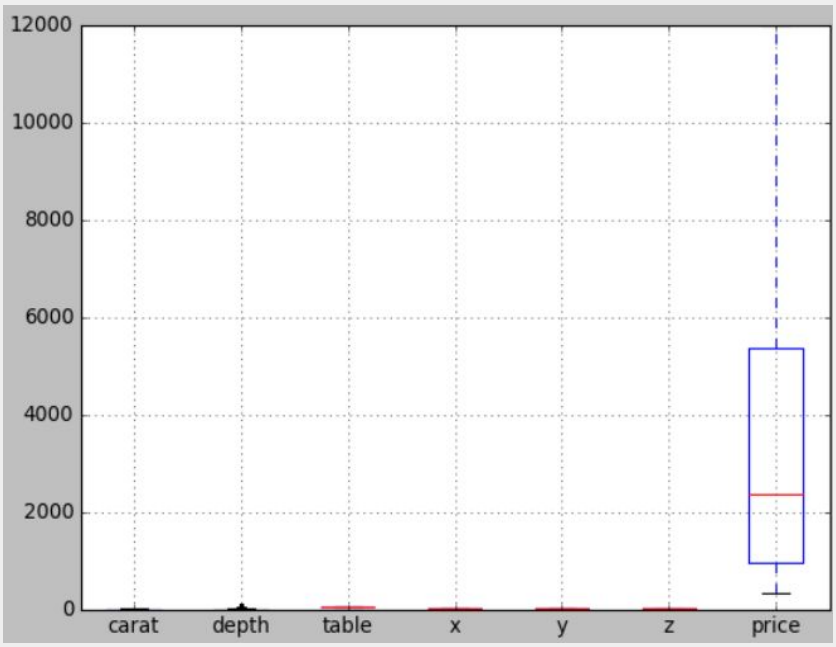
# Univariate analysis

## Histogram



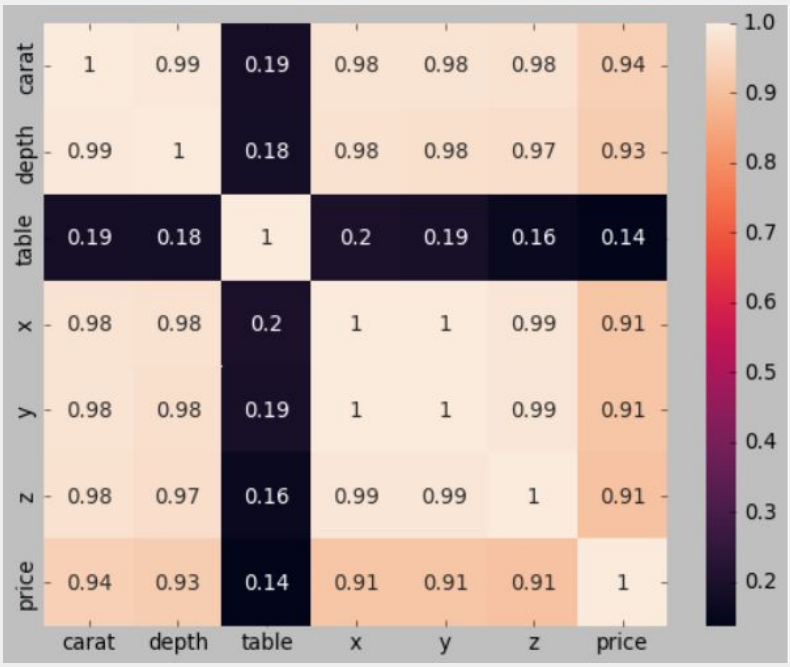

- In the histogram plotted Carat,Depth and Price are skewed to the right, while Table and X have a normal distribution
- Outliers were found and treated for Price variable

## Boxplot




# Bi-variate analysis

## Heatmap



- Pairplot and heat map were drawn for bi-variate analysis
- Depth and Carat had a high correlation, similarly X,Y and Z also had high correlation, thus Depth, Y and Z were dropped to avoid multicollinearity.

1.2 Impute null values if present, also check for the values which are equal to zero. Do they have any meaning or do we need to change them or drop them? Do you think scaling is necessary in this case?

```
In [11]: data.median()

Out[11]: carat       0.70
         depth      61.80
         table      57.00
         x           5.69
         y           5.71
         z           3.52
         price    2375.00
         dtype: float64

In [12]: data.depth = data.fillna(data.median())

In [13]: data.isnull().sum()

Out[13]: carat      0
         cut        0
         color      0
         clarity    0
         depth      0
         table      0
         x          0
         y          0
         z          0
         price      0
         dtype: int64
```

## Scaling

```
In [33]: scaler = preprocessing.StandardScaler()
```

```
In [34]: import warnings
         warnings.filterwarnings("ignore")

         data_train[num_col] = scaler.fit_transform(data_train[num_col])
         data_test[num_col] = scaler.fit_transform(data_test[num_col])
```

- Null values have been imputed using Median
- values equal to Zero are present in the features X,Y and Z, we need not change them as those features have been dropped
- Scaling is necessary due to each feature having values of different units, and has been done using Standard Scaler

1.3 Encode the data (having string values) for Modelling. Data Split: Split the data into test and train (70:30). Apply Linear regression. Performance Metrics: Check the performance of Predictions on Train and Test sets using Rsquare, RMSE.

### Encoding data

```
In [27]: from sklearn import preprocessing
         # label_encoder object knows how to understand word labels.
         label_encoder = preprocessing.LabelEncoder()
         # Encode labels in column 'species'.
         data['cut']= label_encoder.fit_transform(data['cut'])
         data['color'] = label_encoder.fit_transform(data['color'])
         data['clarity'] = label_encoder.fit_transform(data['clarity'])
```

```
In [28]: data.dtypes

Out[28]: carat      float64
         cut          int32
         color        int32
         clarity      int32
         table      float64
         price      float64
```

- Dataset containing string values have been encoded using Label Encoder

# Model Summary for Training Data

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.908
Model:                            OLS   Adj. R-squared:                  0.908
Method:                 Least Squares   F-statistic:                 3.737e+04
Date:                Fri, 17 Apr 2020   Prob (F-statistic):               0.00
Time:                        15:05:51   Log-Likelihood:                -4219.6
No. Observations:               18850   AIC:                             8451.
Df Residuals:                   18844   BIC:                             8498.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1773      0.008    -21.179      0.000      -0.194      -0.161
carat          1.0053      0.002    419.445      0.000       1.001       1.010
cut            0.0208      0.002      9.576      0.000       0.017       0.025
color         -0.0649      0.001    -48.045      0.000      -0.068      -0.062
clarity        0.0764      0.001     58.217      0.000       0.074       0.079
table         -0.0433      0.002    -19.013      0.000      -0.048      -0.039
==============================================================================
Omnibus:                     4266.649   Durbin-Watson:                   1.977
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            17461.296
Skew:                           1.070   Prob(JB):                         0.00
Kurtosis:                       7.202   Cond. No.                         21.7
==============================================================================
```

**RMSE**

```
In [42]: mse = np.mean((regression_model.predict(X_train)-y_train)**2)

In [43]: import math

         math.sqrt(mse)

Out[43]: 0.3026782113470496
```

- the following is the Linear Regression Equation for Train set : **price = -0.18 + 1carat + 0.02cut - 0.06color + 0.08clarity - 0.04table**
- When carat increases by 1 unit, price increases by 1 unit, keeping all other predictors constant. Similarly, when cut increases by 1 unit, price increases by 0.02 units, keeping all other predictors constant.
- There are also some negative coefficient values, for instance, color has its corresponding co-efficient as -0.06. This implies, when the color increases by 1 unit, the price decreases by 0.06 units, keeping all other predictors constant.
- The R-squared and adjusted R-squared are both at 91% implying the model has not been overfitted or underfitted and 91% of the variation in price is explained by the predictors in the model
- The RMSE is at 30% indicating a good fit

## Model Summary for test data

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.903
Model:                            OLS   Adj. R-squared:                  0.903
Method:                 Least Squares   F-statistic:                 1.497e+04
Date:                Fri, 17 Apr 2020   Prob (F-statistic):               0.00
Time:                        15:06:12   Log-Likelihood:                -2053.1
No. Observations:                8079   AIC:                             4118.
Df Residuals:                    8073   BIC:                             4160.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1857      0.013    -13.856      0.000      -0.212      -0.159
carat          1.0022      0.004    265.401      0.000       0.995       1.010
cut            0.0232      0.003      6.669      0.000       0.016       0.030
color         -0.0662      0.002    -30.941      0.000      -0.070      -0.062
clarity        0.0781      0.002     37.943      0.000       0.074       0.082
table         -0.0337      0.004     -9.341      0.000      -0.041      -0.027
==============================================================================
Omnibus:                     1965.996   Durbin-Watson:                   2.031
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             9464.907
Skew:                           1.097   Prob(JB):                         0.00
Kurtosis:                       7.828   Cond. No.                         22.1
==============================================================================
```

**RMSE**

```
In [51]: mse = np.mean((regression_model.predict(X_test)-y_test)**2)

In [52]: import math

         math.sqrt(mse)

Out[52]: 0.3119802194589473
```

- the following is the Linear Regression Equation for Test set : price = -0.185 + 1carat + 0.02cut - 0.07color + 0.07clarity - 0.03table
- When carat increases by 1 unit, price increases by 1 unit, keeping all other predictors constant. Similarly, when cut increases by 1 unit, price increases by 0.02 units, keeping all other predictors constant.
- There are also some negative coefficient values, for instance, color has its corresponding co-efficient as -0.07. This implies, when the color increases by 1 unit, the price decreases by 0.07 units, keeping all other predictors constant.
- The R-squared and adjusted R-squared are both at 90% implying the model has not been overfitted or underfitted and 90% of the variation in price is explained by the predictors in the model
- The RMSE is at 31% indicating a good fit

```
In [54]: from sklearn.feature_selection import RFE
         model = LinearRegression()
         rfe = RFE(model, 1)
         fit = rfe.fit(X_test, y_test)
         print("Num Features: %s" % (fit.n_features_))
         print("Selected Features: %s" % (fit.support_))
         print("Feature Ranking: %s" % (fit.ranking_))

         Num Features: 1
         Selected Features: [False  True False False False False]
         Feature Ranking: [6 1 5 3 2 4]
```

To identify the top five important attributes Feature Selection using Recursive Feature Elimination (RFE) was done and the following are the top five Features:

**1.Carat**

**2.Clarity**

**3.Color**

**4.Table**

**5.Cut**

1.4 Inference: Basis on these predictions, what are the business insights and recommendations.

- The price of the Zirconia can be set based on the Weight of the Carat, higher the Carat higher the prize, similarly for cut quality and Color. If the cut Quality is good increase the price, if the color is J decrease the price. Same applies for Clarity.
- The Zirconia can be segregated based on specific attributes, Eg. Zirconia with high Carat Weight, Premium Cut and D color can be segregated into high profit and Zirconia with low Carat weight, Fair cut and J color can be segregated into low profit
- The top five attributes can be used to deduce the best Zirconia stones from existing ones and arrange them into order of best to worst

# Problem 2

You are hired by a tour and travel agency which deals in selling holiday packages. You are provided details of 872 employees of a company. Among these employees, some opted for the package and some didn't. You have to help the company in predicting whether an employee will opt for the package or not on the basis of the information given in the data set. Also, find out the important factors on the basis of which the company will focus on particular employees to sell their packages.

2.1 Data Ingestion: Read the dataset. Do the descriptive statistics and do null value condition check, write an inference on it. Perform Univariate and Bivariate Analysis. Do exploratory data analysis.

```
In [55]: df = pd.read_csv('Holiday_Package1.csv')
```

```
In [56]: df.head()
```

Out[56]:

| | 0 | Holliday_Package | Salary | age | educ | no_young_children | no_older_children | foreign |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | no | 48412 | 30 | 8 | 1 | 1 | no |
| **1** | 2 | yes | 37207 | 45 | 8 | 0 | 1 | no |
| **2** | 3 | no | 58022 | 46 | 9 | 0 | 0 | no |
| **3** | 4 | no | 66503 | 31 | 11 | 2 | 0 | no |
| **4** | 5 | no | 66734 | 44 | 12 | 0 | 2 | no |

- Dataset has been read and checked using Head() function

Information regarding dataset

```
In [58]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 872 entries, 0 to 871
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Holliday_Package   872 non-null    object
 1   Salary             872 non-null    int64
 2   age                872 non-null    int64
 3   educ               872 non-null    int64
 4   no_young_children  872 non-null    int64
 5   no_older_children  872 non-null    int64
 6   foreign            872 non-null    object
dtypes: int64(5), object(2)
memory usage: 47.8+ KB
```

Null value check and summary of dataset

```
In [109]: df.describe()
```

Out[109]:

| | Holliday_Package | Salary | age | educ | no_young_children | no_older_children | foreign | Prediction |
|---|---|---|---|---|---|---|---|---|
| **count** | 872.000000 | 872.000000 | 872.000000 | 872.000000 | 872.000000 | 872.000000 | 872.000000 | 872.000000 |
| **mean** | 0.459862 | 47729.172018 | 39.955275 | 9.307339 | 0.311927 | 0.982798 | 0.247706 | 0.377294 |
| **std** | 0.498672 | 23418.668531 | 10.551675 | 3.036259 | 0.612870 | 1.086786 | 0.431928 | 0.484987 |
| **min** | 0.000000 | 1322.000000 | 20.000000 | 1.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 |
| **25%** | 0.000000 | 35324.000000 | 32.000000 | 8.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 |
| **50%** | 0.000000 | 41903.500000 | 39.000000 | 9.000000 | 0.000000 | 1.000000 | 0.000000 | 0.000000 |
| **75%** | 1.000000 | 53469.500000 | 48.000000 | 12.000000 | 0.000000 | 2.000000 | 0.000000 | 1.000000 |
| **max** | 1.000000 | 236961.000000 | 62.000000 | 21.000000 | 3.000000 | 6.000000 | 1.000000 | 1.000000 |

```
In [60]: df.isnull().sum()
```

```
Out[60]: Holliday_Package     0
         Salary               0
         age                  0
         educ                 0
         no_young_children    0
         no_older_children    0
         foreign              0
         dtype: int64
```

Duplication check

```
In [61]: dups = df.duplicated()
         print('Number of duplicate rows = %d' % (dups.sum()))
         print(df.shape)

         Number of duplicate rows = 0
         (872, 7)
```

- The Dataset contains 872 observations and 7 features out of which five are integer type and two are object type
- There are no null values or duplicates in the dataset
- 46% of the employees have opted Yes for the Holiday Package while 54% have opted no

Descriptive Analytics

**Mean**

- Salary: 47729.17
- Age: 40
- Educ: 9.3
- no_young_children: 0.3
- no_old_children: 0.9

**Median**

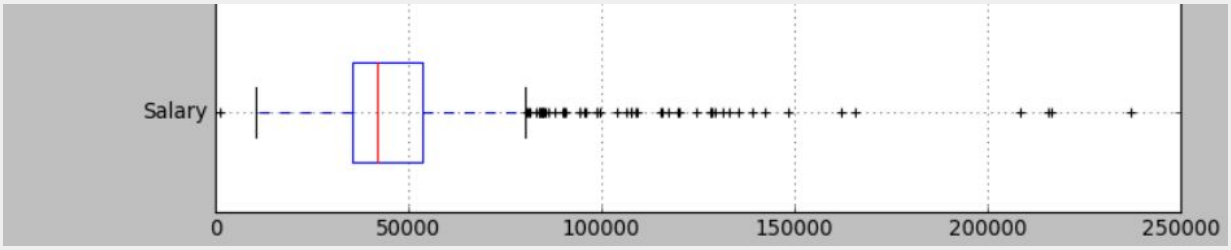
- Salary: 41903.5
- Age: 39
- Educ: 9
- no_young_children: 0
- no_old_children: 1

**Mode**

- Salary: 32197
- Age: 44
- educ: 8
- no_young_children: 0
- no_old_children: 0

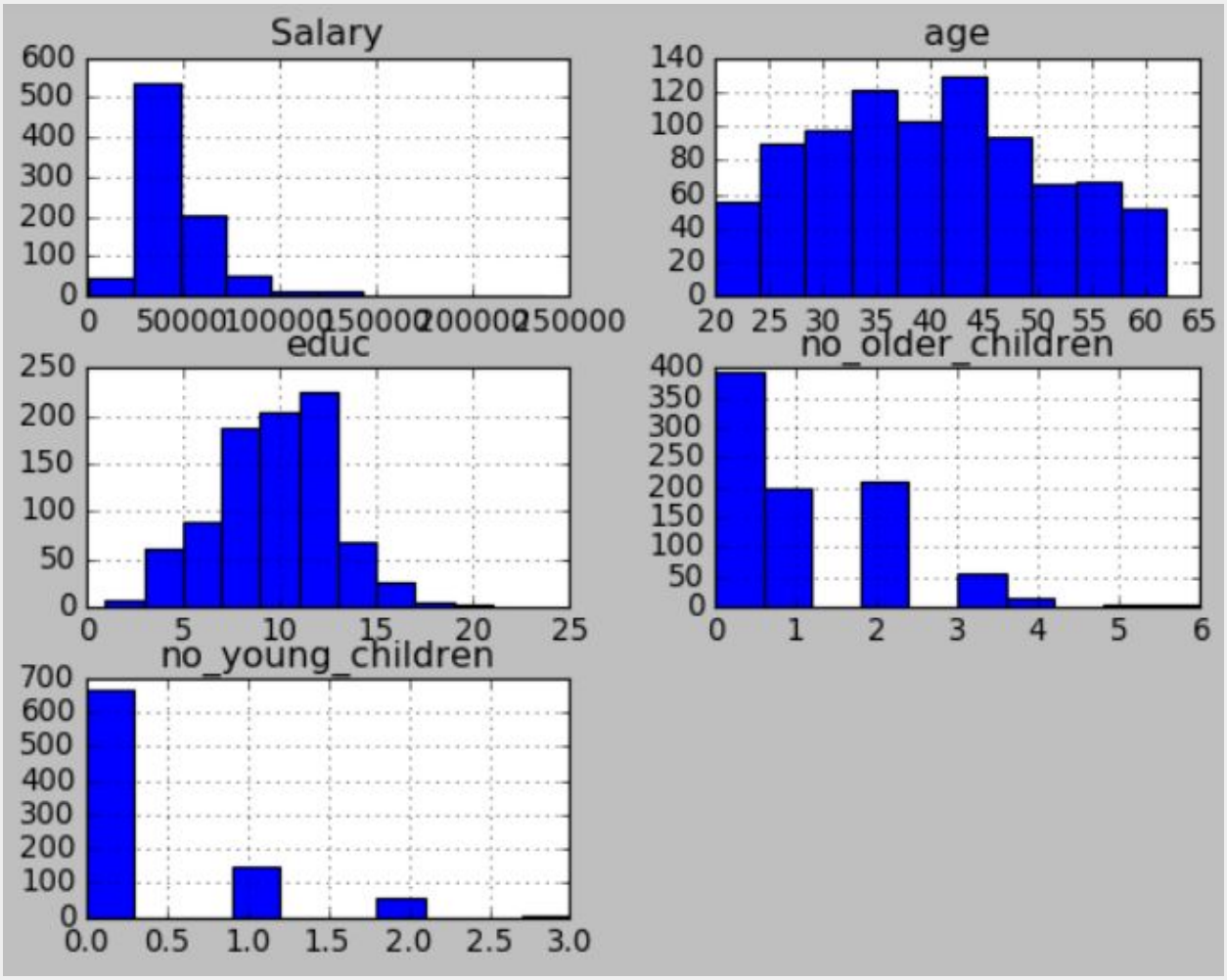
- The most common age of employee is 44
- Most of the employees do not have children, if they do they are all old children
- The salary ranges from 30000 to 50000
- The employees have at least 9 years of education

## Univariate Analysis



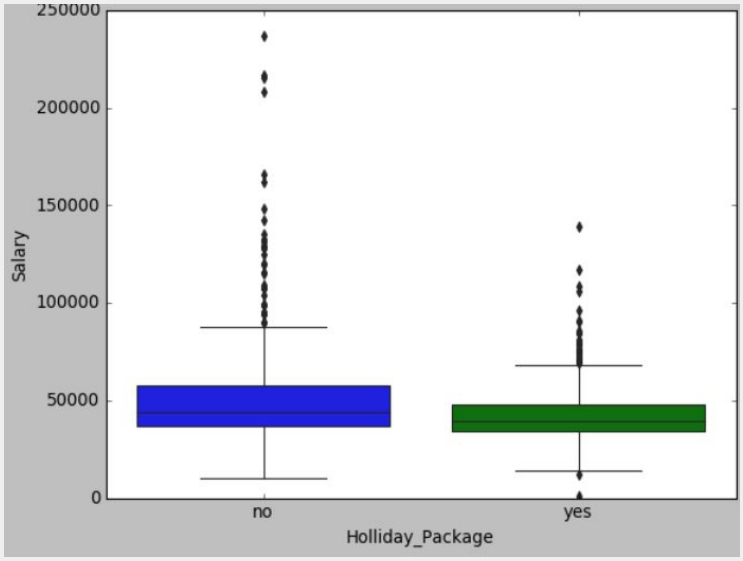

- Although there are outliers present, they are not treated since the Median and mode are fairly close to each other

## Histogram

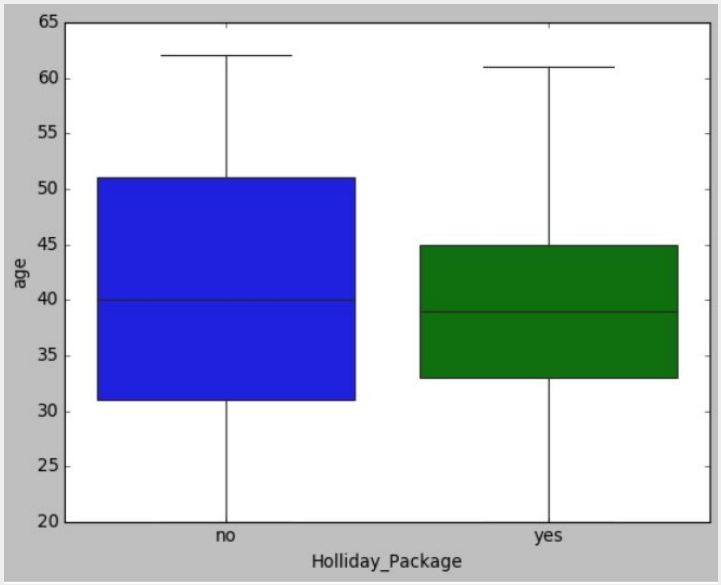



- Salary is rightly skewed
- Age and Educ have a roughly Normal distribution

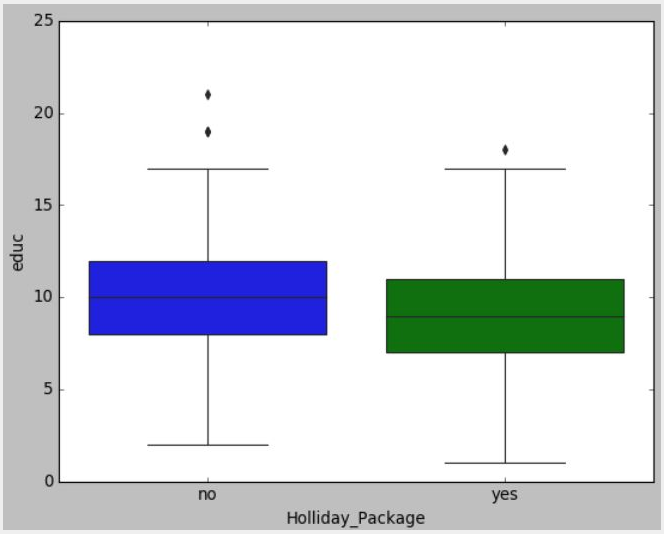
# Bi-variate Analysis



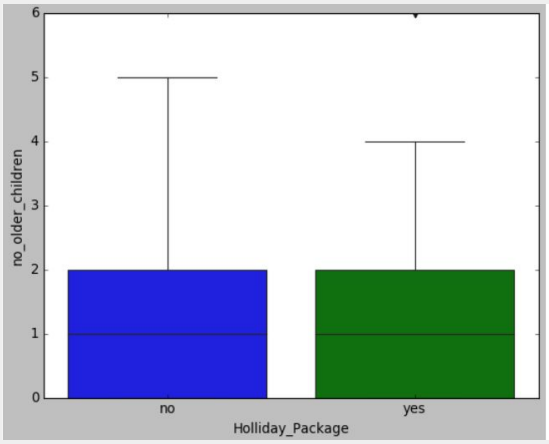

- Employees with higher salary have opted No

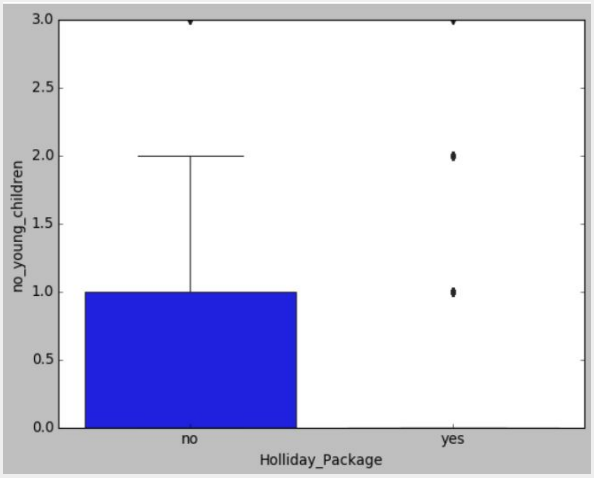



- Age doesn't seem to be affecting Holiday package

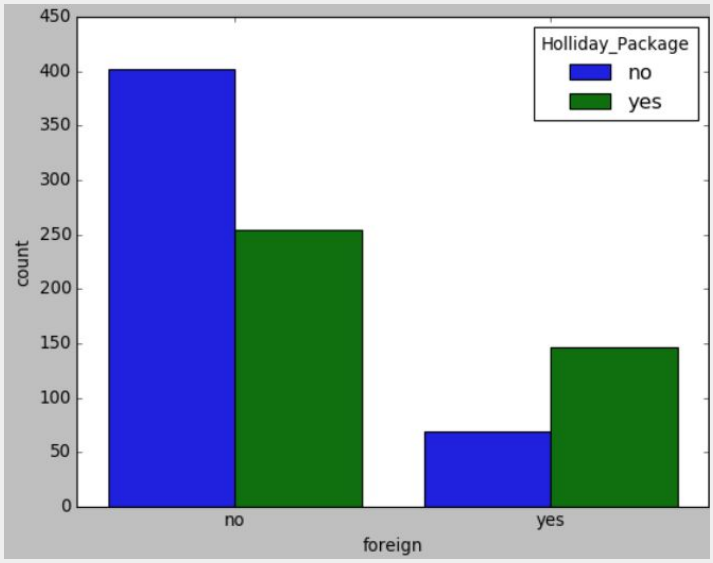



- Education is also not a factor affecting Holiday Package



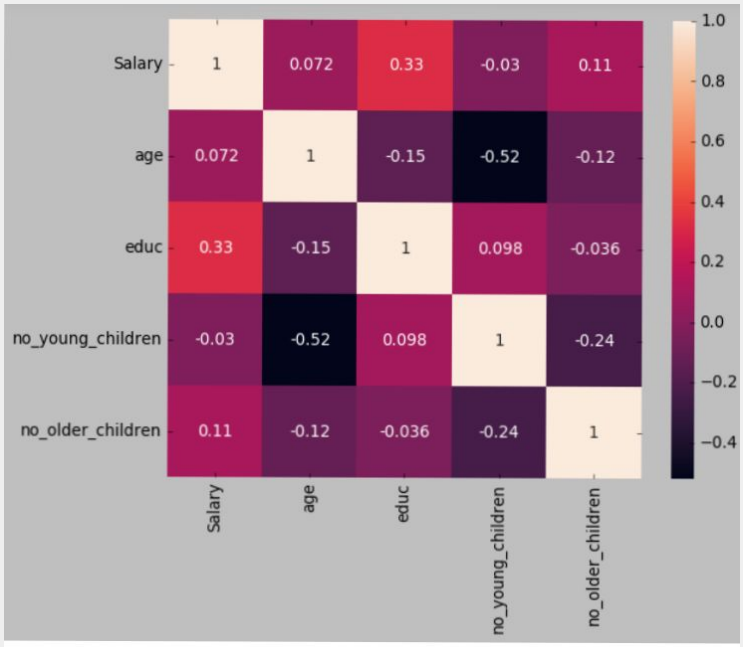




- Both young or old children have no effect on the target variable




- The ratio of non-foreigners are high compared to foreigners
- most of the non Foreigners have opted no while there are some who have opted yes
- Most of the Foreigners have opted yes while the rest has opted no

## Heatmap




- None of the Features have high correlation

2.2 Do not scale the data. Encode the data (having string values) for Modelling. Data Split: Split the data into train and test (70:30). Apply Logistic Regression and LDA (linear discriminant analysis).

```
In [73]: for feature in df.columns:
             if df[feature].dtype == 'object':
                 print('\n')
                 print('feature:',feature)
                 print(pd.Categorical(df[feature].unique()))
                 print(pd.Categorical(df[feature].unique()).codes)
                 df[feature] = pd.Categorical(df[feature]).codes
```

```
feature: Holliday_Package
[no, yes]
Categories (2, object): [no, yes]
[0 1]


feature: foreign
[no, yes]
Categories (2, object): [no, yes]
[0 1]
```

- Data having String Values have been encoded into 0s and 1s wherin no means 0 and Yes means 1

- Logistic Regression and LDA have been applied on the dataset after splitting the data into train and test set in 70:30 ratio

2.3 Performance Metrics: Check the performance of Predictions on Train and Test sets using Accuracy, Confusion Matrix, Plot ROC curve and get ROC_AUC score for each model Final Model: Compare Both the models and write inference which model is best/optimized.

**Logistic Regression**

**Training Data**

- AUC: 58%
- Accuracy:51%
- Precision:42%
- Recall:8%

**Test Data**

- AUC: 63%
- Accuracy:53%
- Precision: 38%
- Recall:9%

Training and Test set results are nearly similar indicating no overfitting or underfitting, overall measures are moderate but not good enough to predict

**Linear Discriminant Analysis**

- Accuracy:66%
- Precision:66%
- Recall:54%

Model Performance is moderate but good enough to predict

Out[94]:

| | lg Train | lg Test | lda |
|---|---|---|---|
| **Accuracy** | 0.52 | 0.53 | 0.66 |
| **Recall** | 0.08 | 0.09 | 0.54 |
| **Precision** | 0.42 | 0.38 | 0.66 |

Upon Comparison LDA is the best optimized model for this dataset as its performance measures are higher than that of Logistic Regression

In [113]:
```
from sklearn.feature_selection import RFE
model = LogisticRegression()
rfe = RFE(model, 1)
fit = rfe.fit(X_train, y_train)
print("Num Features: %s" % (fit.n_features_))
print("Selected Features: %s" % (fit.support_))
print("Feature Ranking: %s" % (fit.ranking_))
```

```
Num Features: 1
Selected Features: [False False False False False  True]
Feature Ranking: [6 4 5 2 3 1]
```

To identify the top five important attributes Feature Selection using Recursive Feature Elimination (RFE) was done and the following are the top five Features:

**1.Foreign**

**2.no_young_children**

**3.no_old_children**

**4.Age**

**5.Educ**

2.4 Inference: Basis on these predictions, what are the insights and recommendations.

- The above model can be used to predict whether an employee will opt for the Holiday Package or not and advertise to them specifically.
- The data can also be used to customize the Holiday package such us destinations which attract children and packages suitable to salaries
- Age can also be used to deduce which type of holiday package and what time of the year employees are highly likely to opt
- More information on Foreign feature can help in further making the Holiday Package attractive
- An important attribute needed for Holiday package prediction is hobbies which is not given here